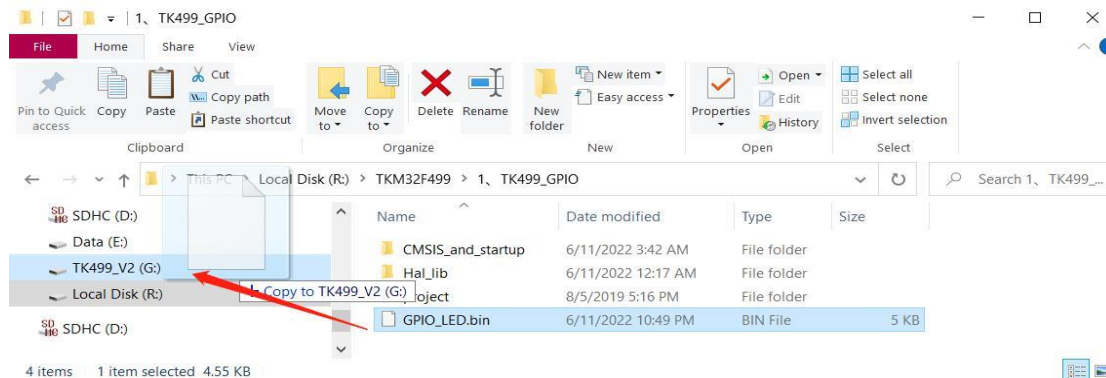


TKM32F499 program and data download method

TKM32F499 supports downloading by dragging and dropping a USB disk, without the need of a downloader, by plugging a USB cable into the computer. Bootloader and application APP are both supported.

Method 1: Download by dragging and dropping from a USB stick

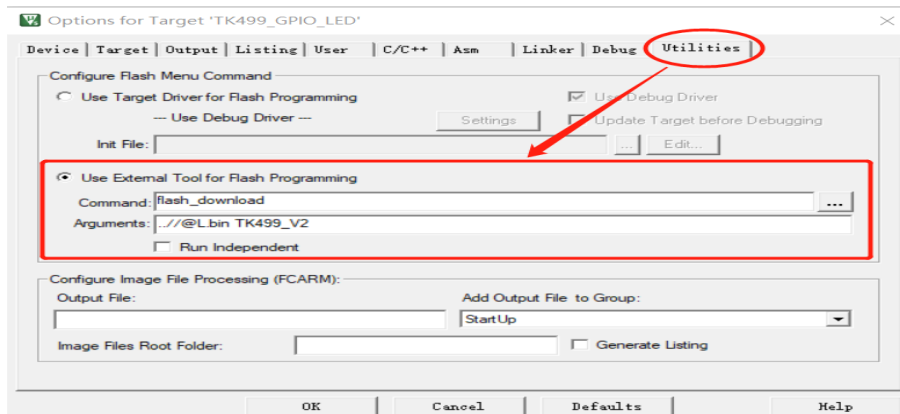
Push up, press reset at the same time, then release the reset and then release the up button, it will enter the program download mode, the computer will pop up a "TK499_V2" disk drive, drag the KEIL project directory generated bin program into the U disk, as follows: the U disk will automatically disappear, if not automatically run. You can press reset, sometimes need to power up and down. (Note: If the pop-up is "TK499" drive letter, it means that there is no bootloader or has been deleted, you need to drag the bootloader. bin in the zip package first, then restart the MCU, and then drag the application APP)



Method 2: Download with KEIL

Push the five-way button upward (press the PA1 button, also known as APP key), press the reset at the same time, then release the reset first, then release the five-way button, and finally click the KEIL  button to download. KEIL will prompt TKM32F499 Download KEIL will prompt TKM32F499 to download , when you hear the "ding-dong" sound of exiting the U disk, it is ready. If it does not run automatically, you can press reset, sometimes you need to power up and down.

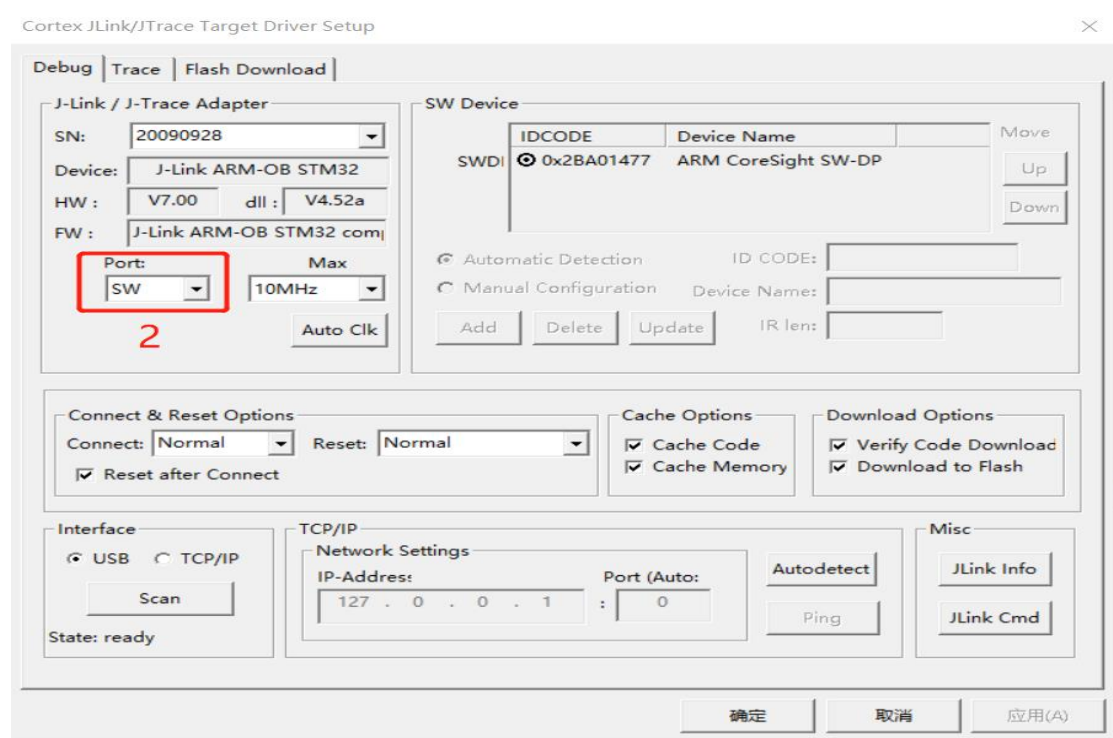
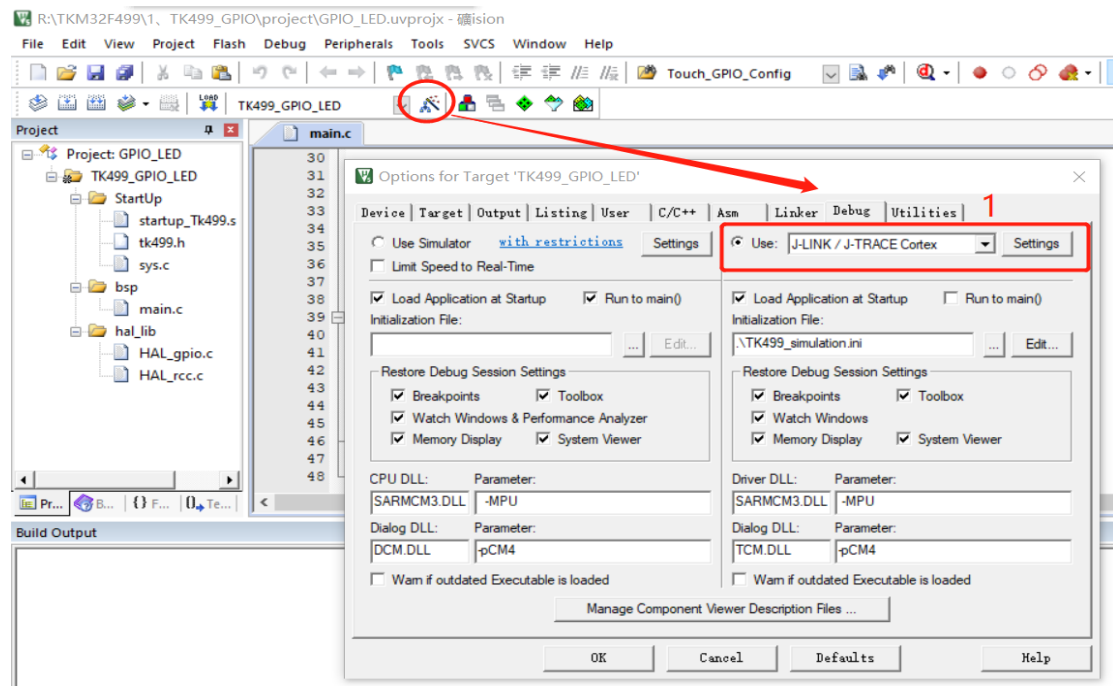
To download with KEIL, the following settings are required flash_download . // @L.bin TK499_V2 (Generally our project is already set up).

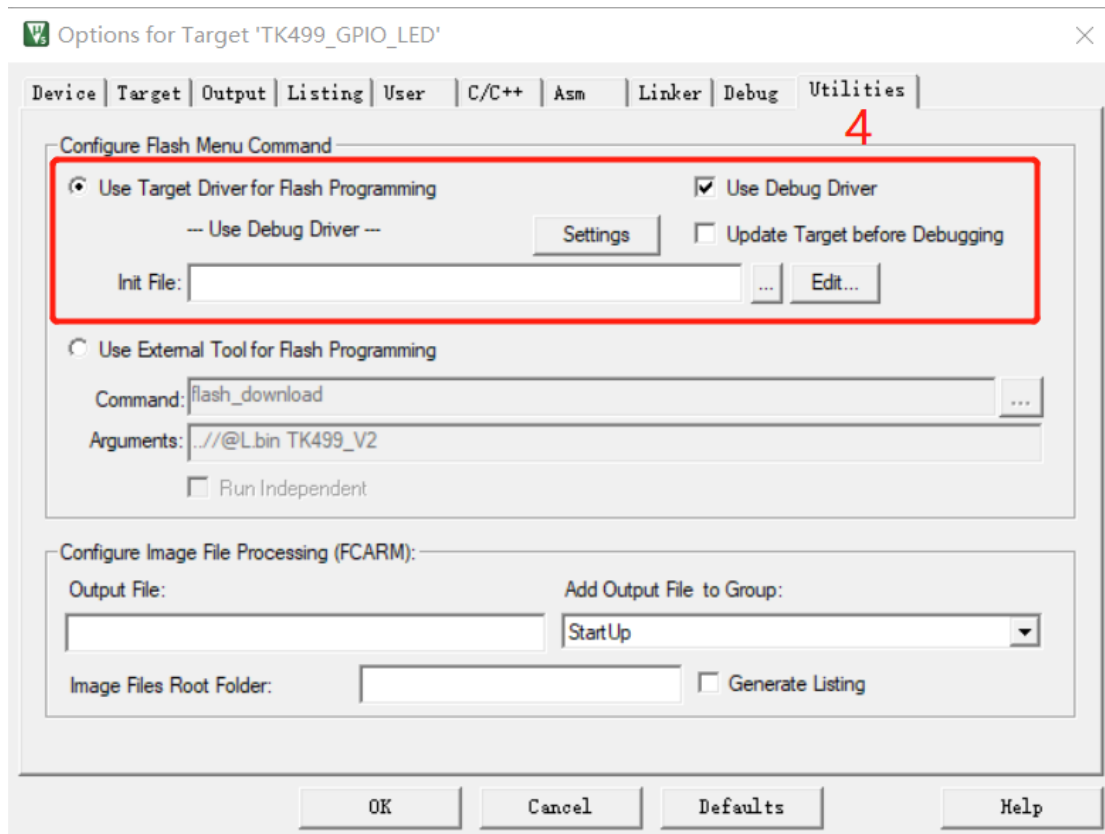
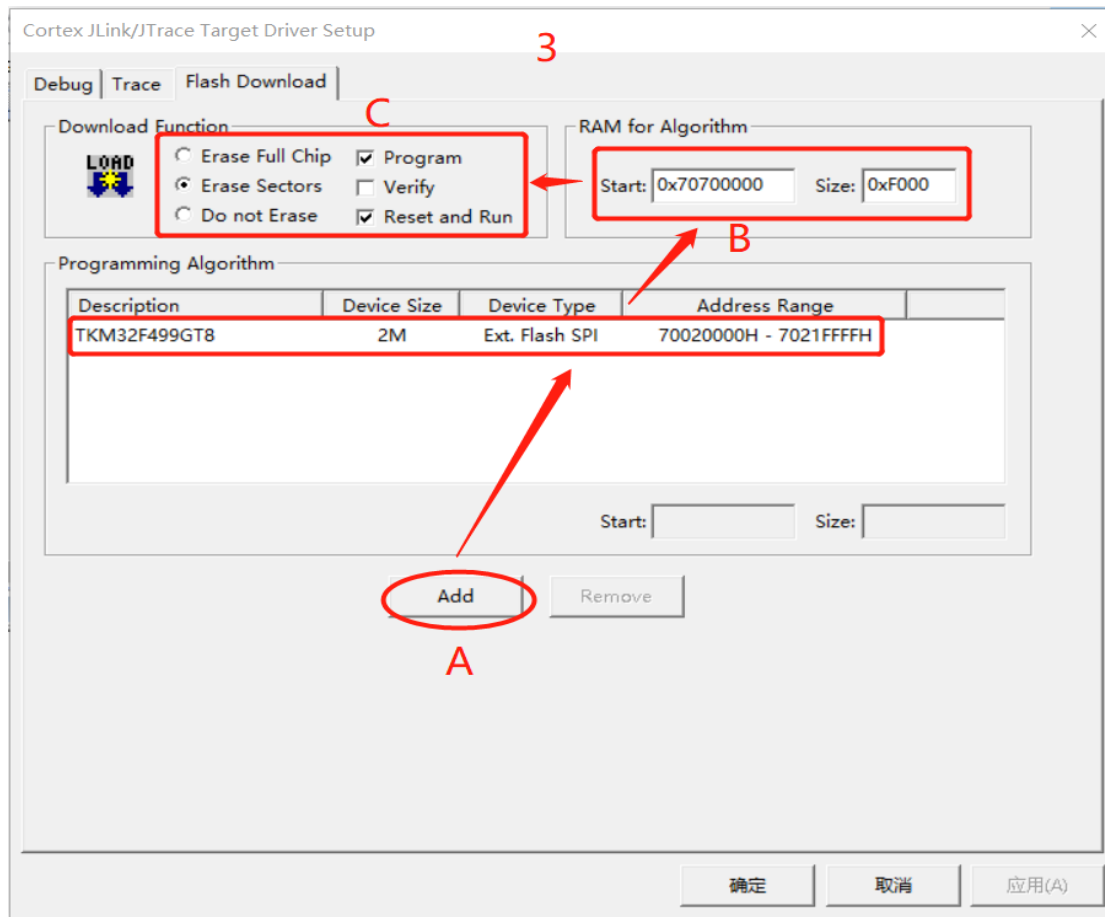


Method 3: Use emulator (e.g. jlink, DAP link) to download

To download with the emulator, you need to download the algorithm. If you are using it for the first time, you need to put the download algorithm "xxxx.FLM" from the "KEIL download algorithm (support all kinds of emulator download)" folder into the installation directory of KEIL\ARM\Flash (for example: C:\Keil_v5\ARM\Flash), and then

make the following settings. After all the settings are done, then click  in the menu bar and you can download it. Microcontroller and emulator wiring method: GND, SWC, SWD

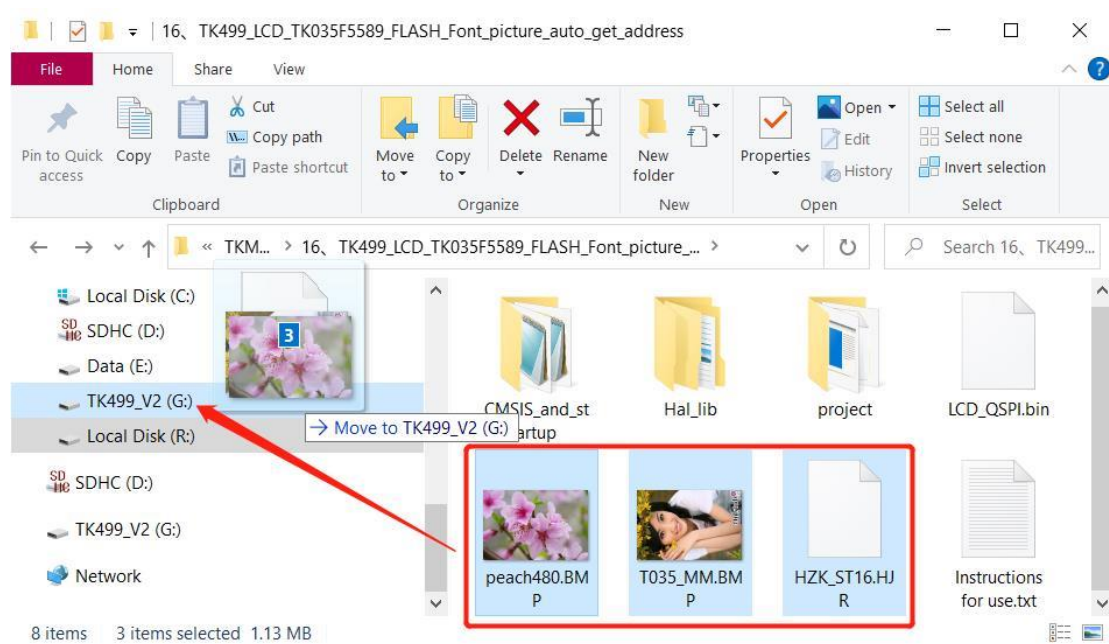




If the chip is used for the first time, or if you need to upgrade the Bootloader, the way to burn the bootloader is: push the five-way button to the left (press the PA13 button, also known as the BOOT key), and then press the reset, then release the reset and then release the up button to enter the Bootloader download mode. At this point, a USB flash drive with the disk character "TK499" will pop up, just drag the Bootloader.bin from the zip package into it. (In general, the development board is already burned Bootloader, **this step can be omitted**)

How can I download data such as font, image and TXT document?

Download data and download the application APP, push up (press the PA1 button, aka APP key), press reset at the same time, then release the reset and then release the up button to bring up a flash drive with the "TK499_V2" drive, drag the files in together.



As shown above, it is the Chinese character library HZK_ST16.HJR and two BMP pictures thrown in together. DEL56789.txt is a special command file for deleting all data, when you throw this file in, the system executes the delete command. The file system currently only supports several formats such as BIN, TXT, JPG, GIF, XBF and HJR for file recognition. If the data you want to download is not in the above format, change the extension to HJR and you can burn it in. The file name currently supports only 8 bytes in length. Because you can burn multiple data at once, the general data is also relatively large, MB level, so the burn time is a little longer than the burn program, to wait until the U disk automatically exit before it is completed, not completed before the power off reset, etc., otherwise the data is not complete. Burn in the first address of the data stored with get_file_address_NOR_FLASH function to get, and then you have the first address can easily read the data. Reference example code: "16、TK499 LCD TK035F5589 FLASH Font picture auto get address. ZIP". To see the real-time burning situation, connect to serial port 1, open the serial assistant on the computer to check the printing information, what file name to upload, how much length, whether the upload is complete are prompted.